D D P M

이건, 주현빈

00. Content tables

Denoising Diffusion Probabilistic Models

Jonathan Ho
UC Berkeley
jonathanho@berkeley.edu

Ajay Jain
UC Berkeley
ajayj@berkeley.edu

Pieter Abbeel
UC Berkeley
pabbeel@cs.berkeley.edu

UC Berkekly

19795회 인용

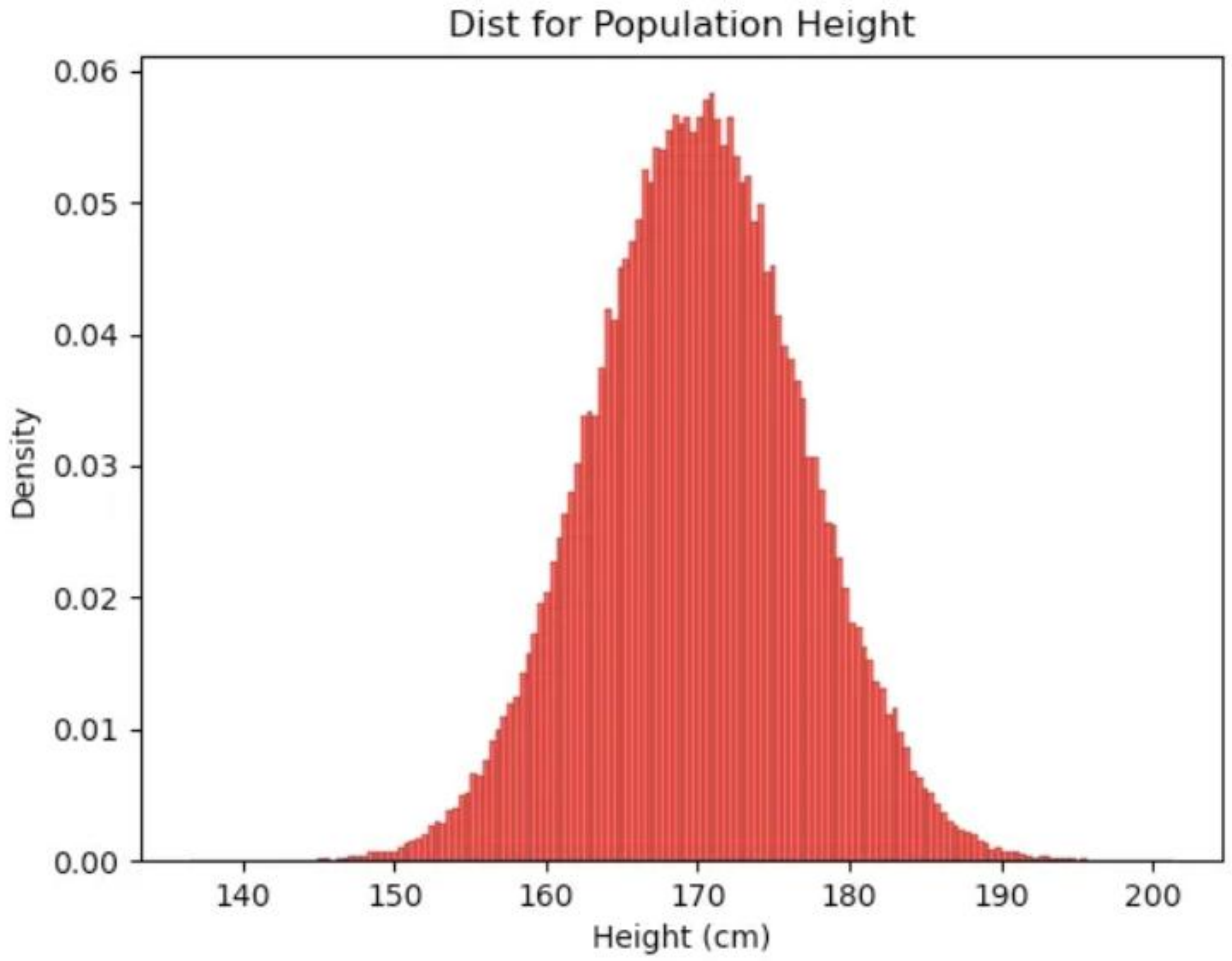
2020 NIPS

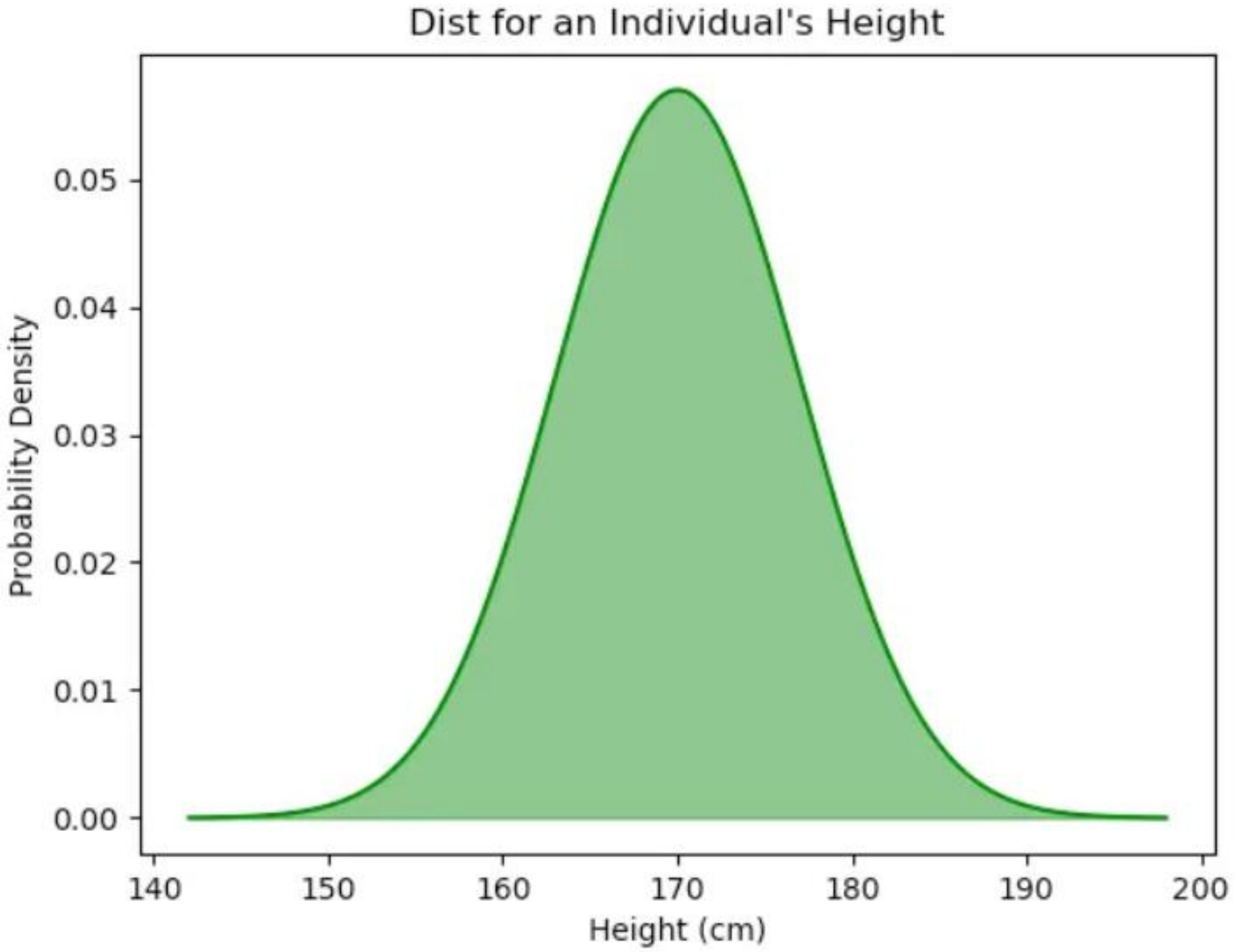
- 이미지 생성 모델 중 하나로, Reverse process를 통해 샘플 생성
- 고품질 이미지 생성 가능 (GAN을 상회)
- Ho et al., 2020이 제안, but, 개념적인 idea는 Sohl-Dickstein et al., 2015 의 diffusion에서 차용

1. 생성 모델이란?

-> 생성 모델은, '그럴듯' 한 데이터를 생성하는 것을 목표로 함

가장 단순한 생성 모델의 예시를 들어보자.

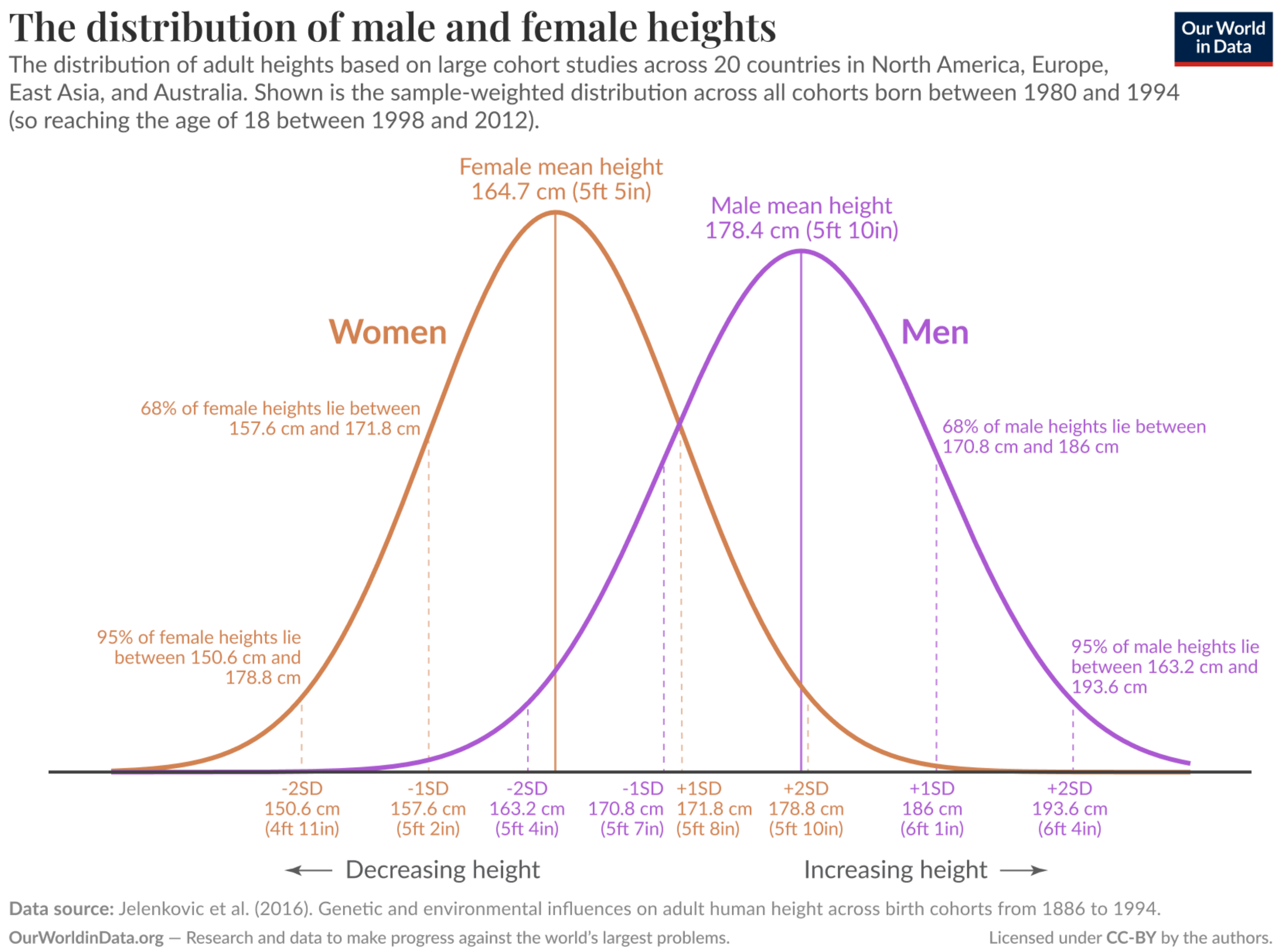




- 사람의 키 데이터와 같은 1차원 scalar data를 생성하는 generative model을 학습시킨다고 해보자.
- 우측과 같은 1dim gaussian distribution의 모수를 찾으면 됨.
- 데이터가 gaussian을 따른다면, 그냥 평균과 표준편차를 구하면 그 모델이 가장 최적의 모델일 것
- 새로운 데이터를 생성한다면, 우측의 정규 분포 곡선에서 샘플링 하면 됨

근데, 분포가 조금이라도 복잡해지면?

ex) 남자와 여자의 키 분포



- 위의 데이터를 1 dim guassian model로 모델링 할 수 있는가? A) 못한다...
- 그러면 어떻게? ... deep learning을 활용하자 ! -> 아까는 최적의 모델을 찾기 위해서 평균과 표준편차를 구했다.
- 딥러닝에서는 어떻게 해야 하는가? A) MLE

2. KL Divergence

- 두 확률 분포의 차이를 정량적으로 구하는 법
- Cross Entropy와도 유사하다 (사실, classification task의 CE loss는 KL divergence의 특수 case)

$$H(P) = -\sum_x P(x) \log P(x)$$

- 확률분포 P의 Entropy

$$H(P, Q) = -\sum_x P(x) \log Q(x)$$

- 확률분포 P와 Q의 Cross Entropy

$$D_{KL}(P \parallel Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)}$$

- KL Divergence (P, Q의 CE – P의 E)

$$D_{KL}(p \parallel q) = \int p(x) \log \frac{p(x)}{q(x)}\, dx$$

- KL divergence 식의 연속형

ex. Classification에서,

true label = [1, 0, 0]이고,
prediction = [0.7, 0.2, 0.1]일 때

- $H(P) = 0$ (확실한 정보니까 불확실성 없음)
- $H(P, Q) = -\log(0.7) \approx 0.357$
- $D_{KL}(P \parallel Q) = H(P, Q) - H(P) = 0.357 - 0 = 0.357$

-> KL divergence를 loss에 쓰겠다는 것은, 두 확률 분포를 비슷하게 만들도록 파라미터를 학습하겠다는 뜻과 동일

외워야 할 것 (KL divergence의 성질)

- KL divergence의 최솟값은 0 (완전히 같을 때)
- KL divergence는 0 이상의 값을 가짐

3. Maximum Likelihood Estimation (MLE)

앞서, 생성 모델은 그럴듯한 데이터를 생성하는 것이 목표라고 하였음

-> 그럴듯한하다는게 수학적으로 어떻게 표현되는가? -> Maximum Likelihood Estimation

MLE : 관측 데이터를 바탕으로, 가장 타당한 모수를 추정하는 것이 Maximum Likelihood Estimation

ex) 승/패가 있는 룰렛의 승률을 모를 때, 10판을 진행해서 7판을 이겼다. 룰렛의 승률은 몇 %로 추산하는가?

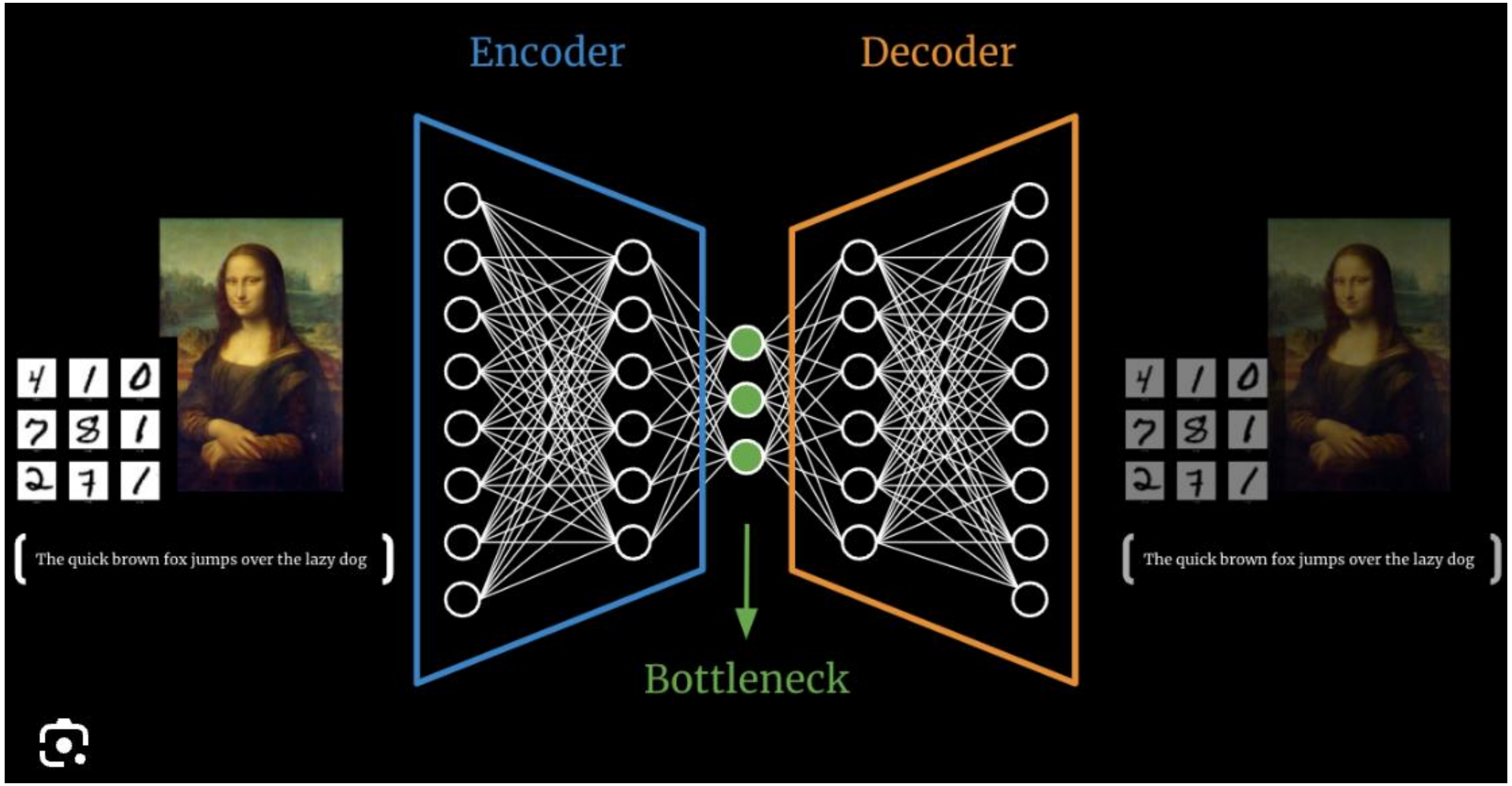
룰렛의 승률을 theta라고 할 때, 해당 관측의 likelihood는 아래의 식이다.

$$p_\theta(x_1, ..., x_{10}) = \prod_{i=1}^{10} p_\theta(x_i) = \theta^{\sum_{i=1}^{10} x_i}(1-\theta)^{10-\sum_{i=1}^{10} x_i}$$

위 공식의 값을 최대화하는 theta가, 가장 그럴 듯 한 theta이다. 딥러닝에서는 theta = learnable parameter로 정의됨

4. Latent Variable (z)

- 관찰할 수 없지만, 데이터 생성에 영향을 주는 숨겨진 요인
- 고차원 데이터를 저차원으로 압축하여 표현 -> 단순한 픽셀의 나열인 원본 x보다, latent variable z는 semantic함



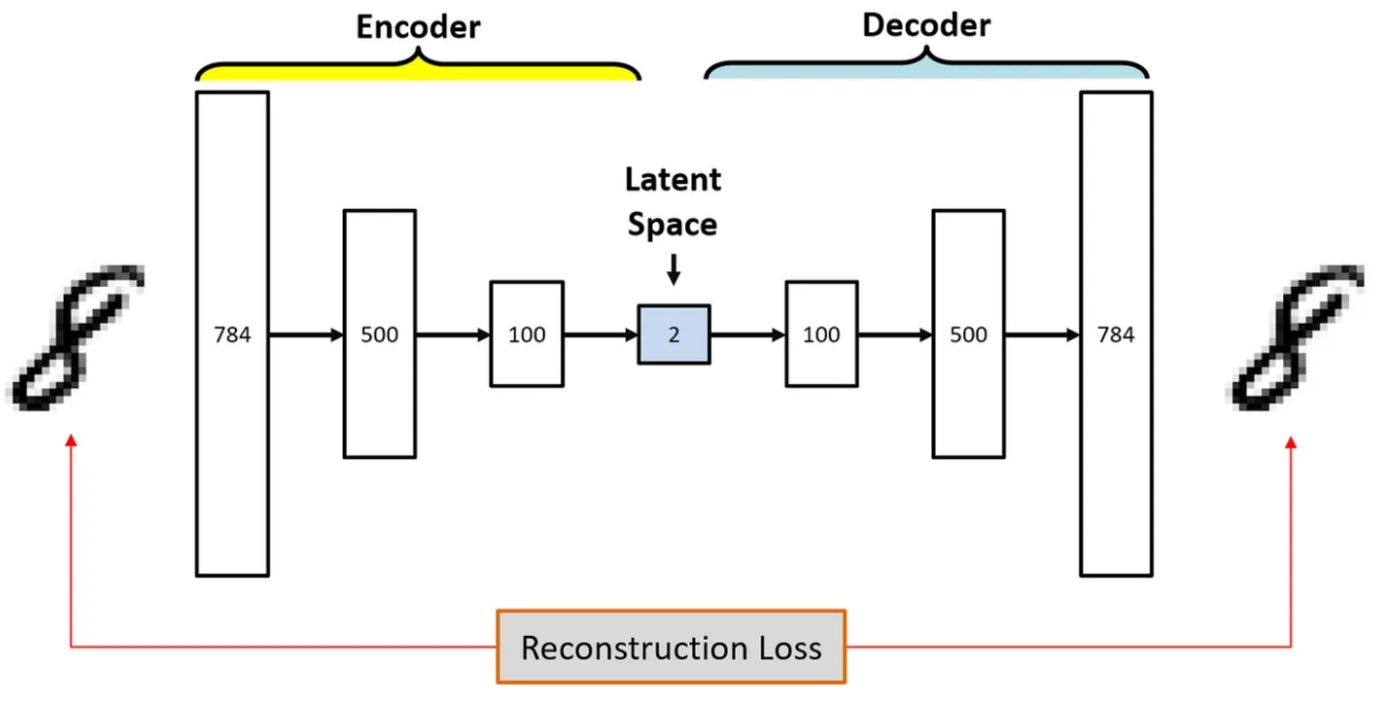
cf) Autoencoder

Autoencoder는 input image x를,
압축했다가 복원하는 식으로 학습하여,
(MSE loss)

정보의 효율적인 압축인 Encoder를 활용

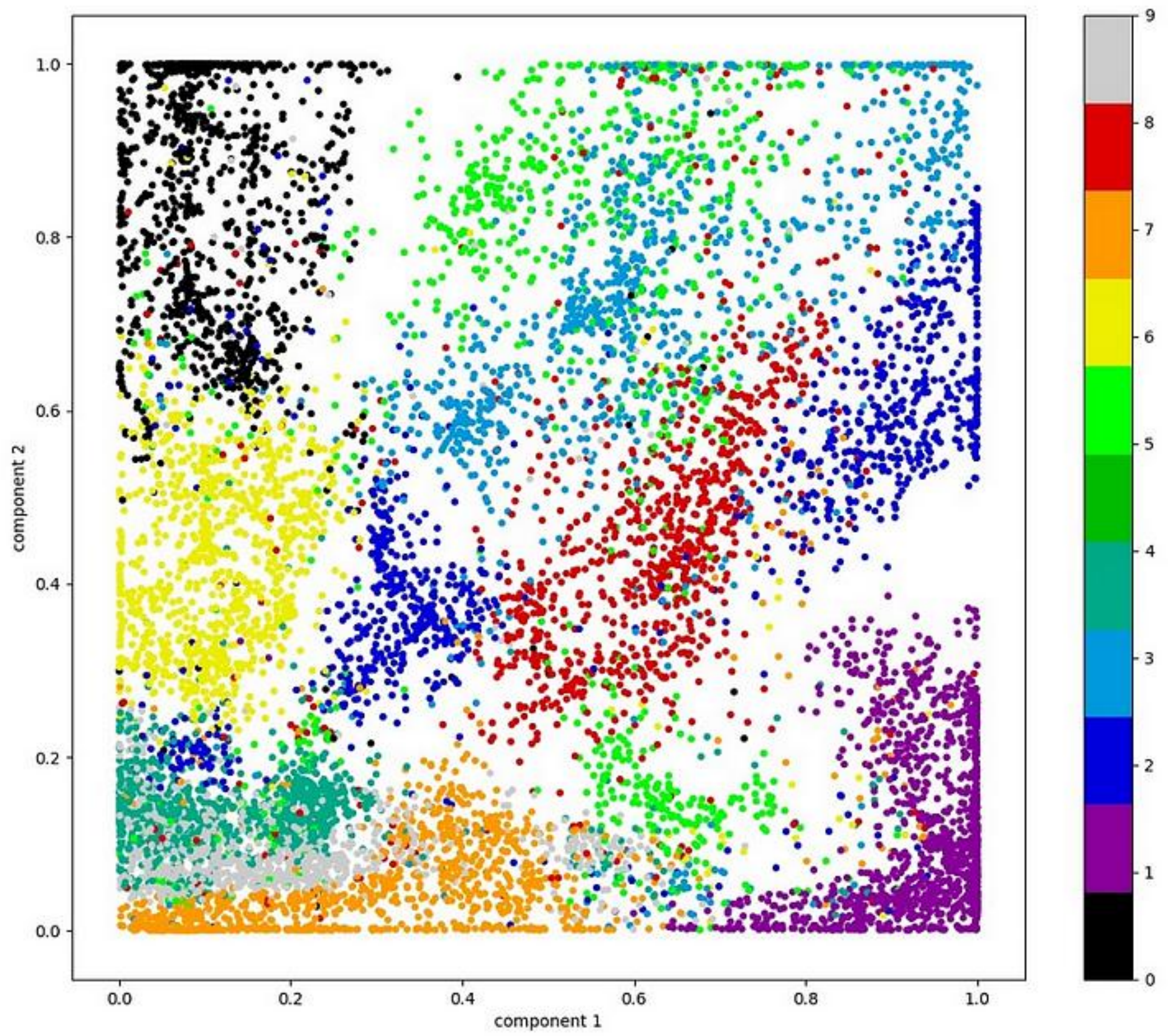
1. Autoencoder의 한계점

- latent variable의 dim = 2로 한 MNIST dataset으로 학습한 autoencoder가 있을 때, 잠재 공간의 분포를



Autoencoder Architecture (Image by the author)

Examples of Image Generation by Autoencoder (Image by the author)



1. 잠재 공간이 이산적

2. 잠재 공간의 해석이 어려움

3. 생성된 이미지의 품질이 낮다

2. 생성 모델로서의 VAE

- 앞선 AE의 한계

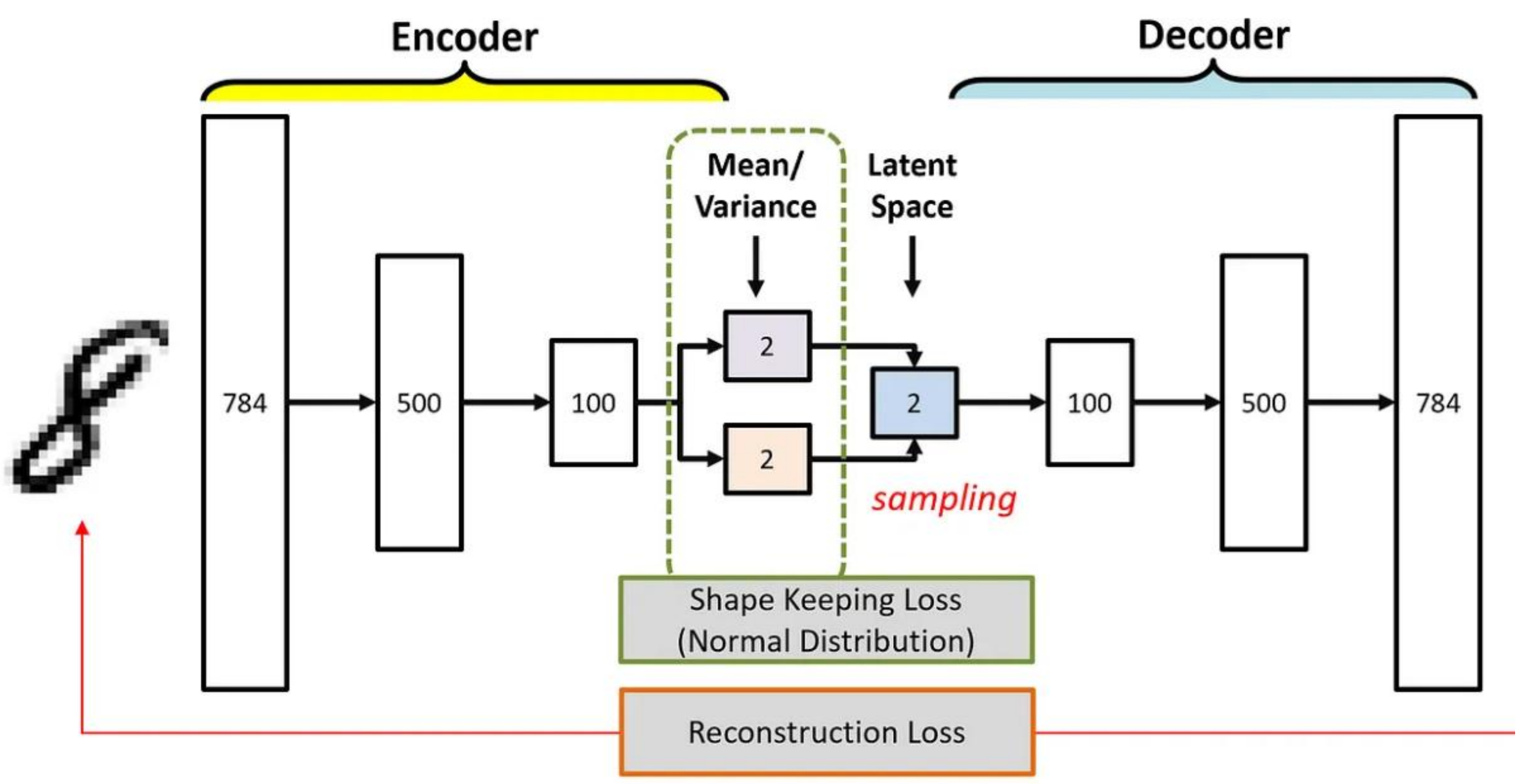
1. 잠재 공간이 이산적

를 극복하기 위해서는, x라는 데이터로부터 압축된 z가 한 점이면 안됨 -> 확률 분포로 만들자 !

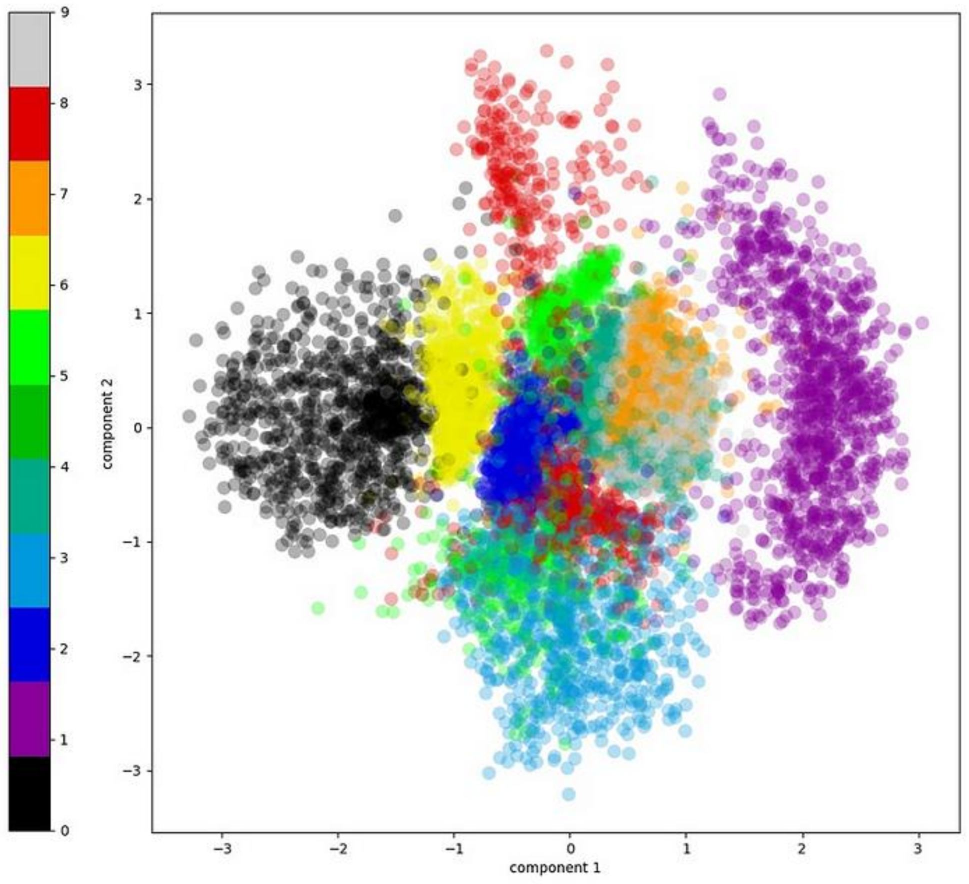
2. 잠재 공간의 해석이 어려움

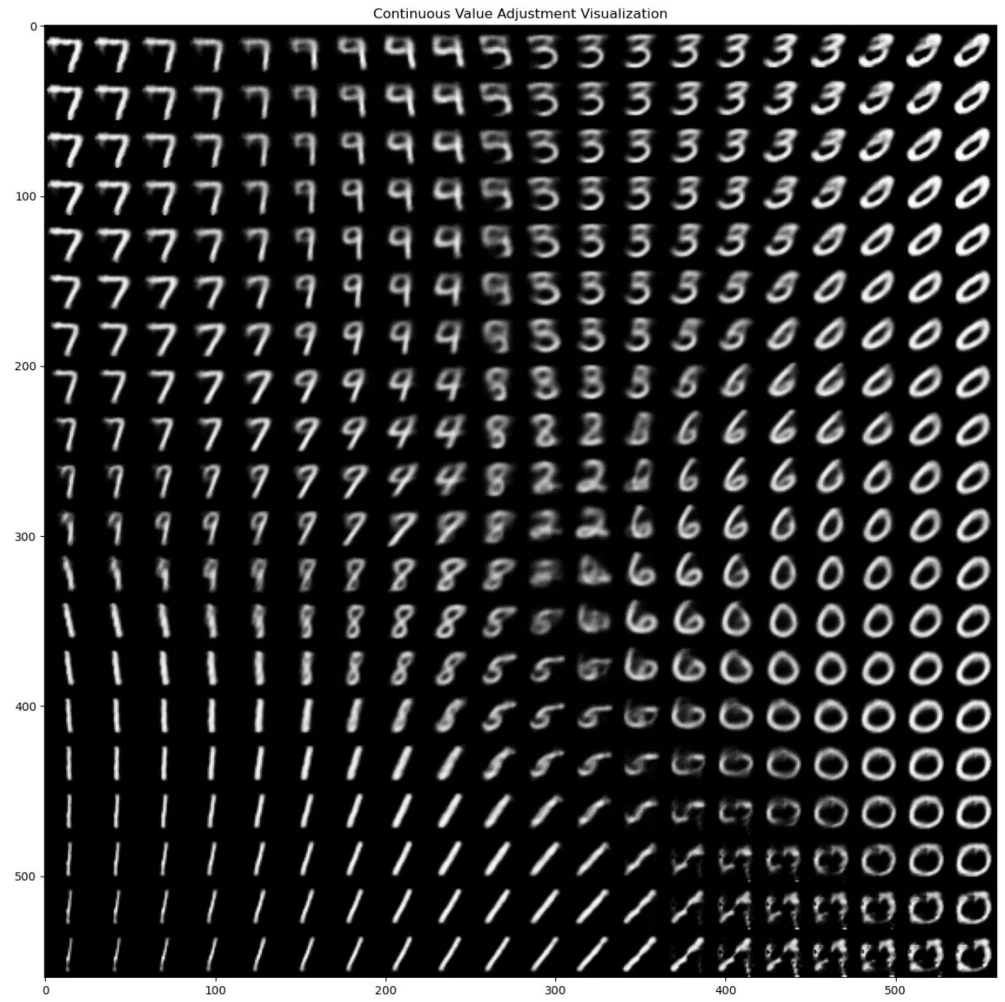
-> 잠재 공간을 보다 정렬될 수 있도록, 확률 분포를 원점에서 크게 벗어나지 않게 하자

3. 생성된 이미지의 품질은, 위의 두 문제를 해결하여 좋아짐 !



Variational Autoencoder Architecture (Image by the author)





3. MLE를 활용한 Loss term의 유도와, KL divergence

- 생성 모델의 목표 -> MLE

$$\log p(x) = \log \int p(x, z)\, dz$$

Eq. 1- maximize logp(x) -> maximize하는 theta 찾기

$$p(x, z) = p(x|z)p(z)$$

Eq. 2-조건부 확률 정의

$$\log p(x) = \log \int q(z|x) \frac{p(x|z)p(z)}{q(z|x)}\, dz$$

Eq. 3-분모분자에 q(z|x)를 곱함

$$\log p(x) \geq \int q(z|x) \log \frac{p(x|z)p(z)}{q(z|x)}\, dz$$

Eq. 4-옌센 방정식 - intractable한 p(z|x)를 근사하는, encoder q(z|x)

$$\mathbb{E}_{q(z|x)} [\log p(x|z)] - \mathrm{KL}(q(z|x) \| p(z))$$

$$\mathcal{L}(x) = -\mathbb{E}_{q(z|x)} [\log p(x|z)] + \mathrm{KL}(q(z|x) \| p(z))$$

4. VAE를 통해서 얻은 insight

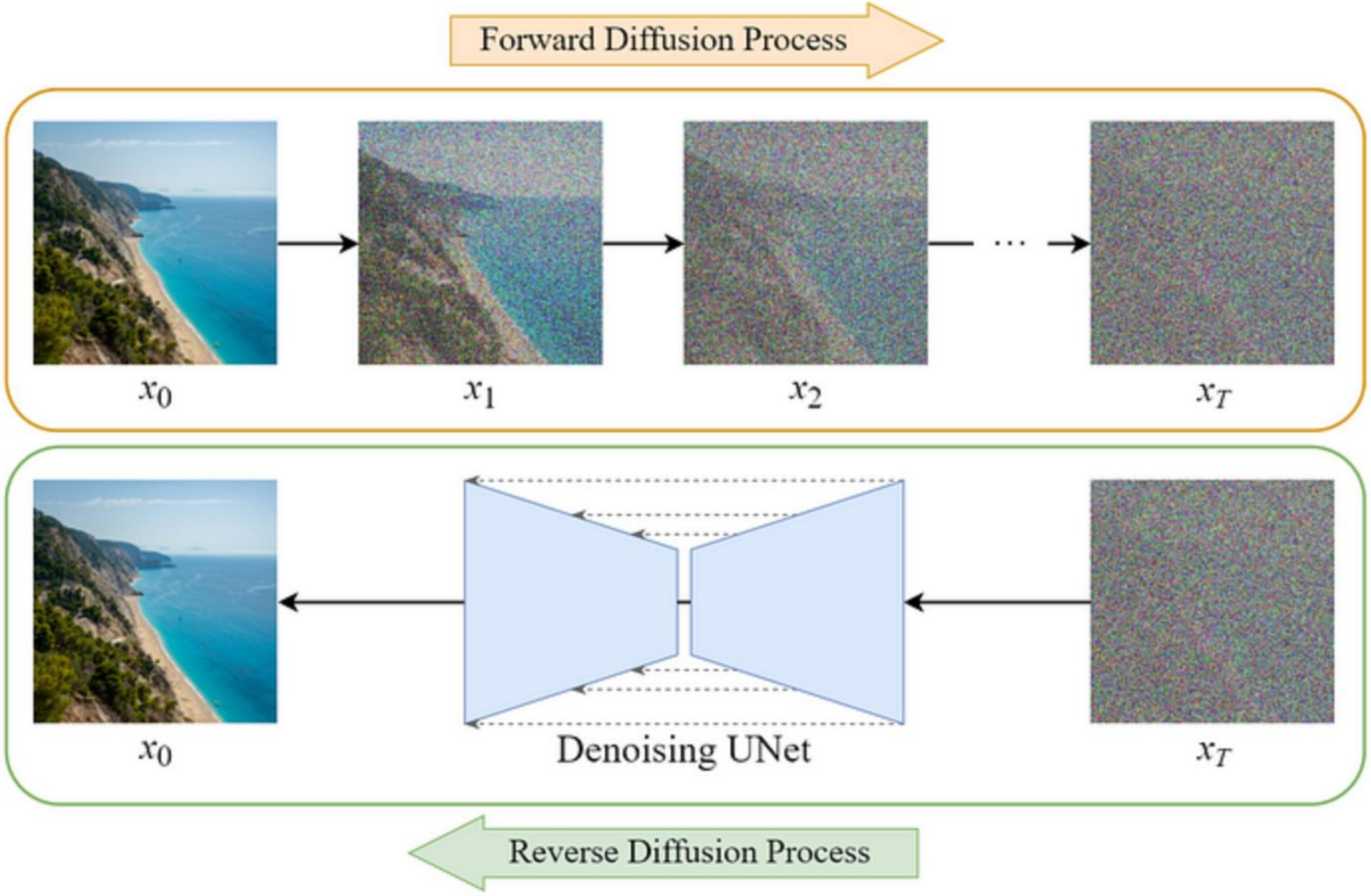
- 새로운 이미지를 생성하기 위해서는, latent space가 continuous 해야 함 -> 이를 위한 방법이, x를 확률 변수로 매핑하는 것

- 일반적으로, 이러한 확률 변수 매핑 과정은, Guassian을 활용한다

- 생성 모델의 Loss 도출 과정은, MLE를 따른다, 모델의 구조마다 식은 다르겠지만, 근본은 MLE이다

1. DDPM base architecture

- Forward diffusion process, Backward diffusion process가 존재
- Backward diffusion process (Reverse diffusion process)를 통해 이미지 생성



Diffusion / VAE의 차이점

1. latent variable z가 존재하지 않음

2. shape의 변화가 없음 (UNet 활용)

3. Learnable Parameter는 오직 UNet

Denoising UNet은, noise로 오염된 image로부터 noise를 예측한다 !

완전히 noisy한 Image x_T로부터, noise 제거 과정을 T번 거쳐서 이미지를 생성함

2. Forward diffusion process

- Markov Chain 기반의 점진적 noise 추가 (x_0 = 원본 이미지, T = 총 Timestep)
- 각 step은 이전의 step에만 영향을 받음을 가정 -> Markov Property
- Image는 결국 순수 노이즈 x_T가 된다
- 사전에 고정된 방식으로 정의 (Beta_t, Alpha_t - hyperparameter)

$$q(x_{1:T}|x_0) = \prod_{t=1}^{T} q(x_t|x_{t-1})$$

- Eq. 1 - markov property

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}\, x_{t-1}, \beta_t \mathbf{I})$$

Eq. 2 – forward process (점화적 식)

$$x_t = \sqrt{1-\beta_t} \cdot x_{t-1} + \sqrt{\beta_t} \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I})$$

Eq. 3 – forward process (수식적 이해)

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon$$

Eq. 4 – forward process (closed form)

식이 어려우면

1. forward는 고정된 방식으로 정의

2. x_t를 구하는 것은 x_0로부터 한번에 구해짐

3. forward는 stochastic하다

-> 이정도만 알고 가자!

2. Backward diffusion process

- Markov Chain 기반의 점진적 noise 제거 (x_0 = 원본 이미지, T = 총 Timestep)
- 각 step은 이전의 step에만 영향을 받음을 가정 -> Markov Property
- 순수 Noise x_T로부터, backward diffusion process를 반복하면 새로운 이미지를 생성해낼 수 있다
- UNet은 x_t와, t를 받아서, noise (epsilon)을 출력한다

$$p_\theta(x_{0:T}) = p(x_T)\prod_{t=1}^{T} p_\theta(x_{t-1}|x_t)$$

- Eq. 1 - markov property

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}t}} \cdot \epsilon\theta(x_t, t)\right)$$

Eq. 2 – 모델이 예측한 epsilon으로, x_t-1을
deterministic하게 구하기 (앞 페이지 Eq. 3참고)

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_t)$$

Eq. 3 – backward process (점화적 식)

$$x_{t-1} = \mu_\theta(x_t, t) + \sigma_t \cdot z, \quad z \sim \mathcal{N}(0, I)$$

Eq 4. – 샘플링 식

식이 어려우면

1. alpha, alpha-bar, beta는 상수이다

2. model(UNet)은, x_t, t를 입력받아,
x_t에 더해진 noise인 epsilon을 반환

3. x_t와, epsilon으로부터 x_t-1을 구할
수 있는데, 또 확률적으로 구한다
(continous latent space)

-> 이정도만 알고 가자!

05. Loss Function (Denoising Diffusion Probabilistic Models)

Loss Function (손실함수)

- **모델의 예측값**과 **실제 정답값** 사이의 차이(오차)를 수치로 나타낸 함수
  (이 값을 최소화 하는 것이 모델 학습의 목표)

Probability Distribution(확률분포)

- 확률 분포(Probability Distribution)는 어떤 일이 일어날 가능성(확률)을 **수학적으로 표현한 것**

Diffusion model 관점에서 Loss Function (손실함수)

- 모델의 확률분포 pθ(backward) 를 데이터에서 유도된 확률분포 q(forward)에 가깝게 만들자
  (pθ≈q 즉, 모델이 진짜 노이징/복원 과정을 **가장 그럴듯하게 흉내 내도록 학습)**

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t} \cdot x_{t-1}, \beta_t I) \longleftarrow p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_t)$$

forward　　　　　　　　　　backward

05. Loss Function (Denoising Diffusion Probabilistic Models)

Diffusion Model Loss(2015)

$$\mathbb{E}_q\Big[\underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0)\,\|\,p_\theta(\mathbf{x}_T))}_{L_T} + \sum_{t=2}\underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)\,\|\,p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}}\underbrace{-\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)}_{L_0}\Big]$$

유도

Diffusion Model Probabilistic Loss(2020)

$$\mathbb{E}_{t,\mathbf{x}_0,\epsilon}\left[\left\|\epsilon - \epsilon_\theta\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t\right)\right\|^2\right] \quad , \ \epsilon \sim \mathcal{N}(\mathbf{0},\mathbf{I})$$

05. Loss Function (Denoising Diffusion Probabilistic Models)

Diffusion Model Loss(2015)

$$\mathbb{E}_q\left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right]$$

Diffusion Model Loss(2015)

$$\mathbb{E}_q\left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right] = \mathbb{E}_{q(\mathbf{x}_{0:T})}\left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})}\right]$$

05. Loss Function (Denoising Diffusion Probabilistic Models)

Diffusion Model Loss(2015)

$$\mathbb{E}_q\left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right] = \mathbb{E}_{q(\mathbf{x}_{0:T})}\left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})}\right] = \mathbb{E}_q\left[\log \frac{\prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_T)\prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}\right]$$

Diffusion Model Loss(2015)

$$\mathbb{E}_q\left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right] = \mathbb{E}_{q(\mathbf{x}_{0:T})}\left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})}\right] = \mathbb{E}_q\left[\log \frac{\prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_T)\prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}\right]$$

$$= \mathbb{E}_q\left[\log \frac{\prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_T)\prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}\right]$$

Diffusion Model Loss(2015)

$$\mathbb{E}_q\left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right] = \mathbb{E}_{q(\mathbf{x}_{0:T})}\left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})}\right] = \mathbb{E}_q\left[\log \frac{\prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_T)\prod_{t=1}^T p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}\right]$$

$$= \mathbb{E}_q\left[\log \frac{\prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_T)\prod_{t=1}^T p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}\right] = \mathbb{E}_q\left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=1}^T \log \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}\right]$$

Diffusion Model Loss(2015)

$$\mathbb{E}_q\left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right] = \mathbb{E}_{q(\mathbf{x}_{0:T})}\left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})}\right] = \mathbb{E}_q\left[\log \frac{\prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_T)\prod_{t=1}^T p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}\right]$$

$$= \mathbb{E}_q\left[\log \frac{\prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_T)\prod_{t=1}^T p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}\right] = \mathbb{E}_q\left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=1}^T \log \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}\right]$$

$$= \mathbb{E}_q\left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^T \log \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}\right]$$

05. Loss Function (Denoising Diffusion Probabilistic Models)

Diffusion Model Loss(2015)

$$\mathbb{E}_q\left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right] = \mathbb{E}_{q(\mathbf{x}_{0:T})}\left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})}\right] = \mathbb{E}_q\left[\log \frac{\prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_T)\prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}\right]$$

$$= \mathbb{E}_q\left[\log \frac{\prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_T)\prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}\right] = \mathbb{E}_q\left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=1}^{T} \log \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}\right]$$

$$= \mathbb{E}_q\left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^{T} \log \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}\right]$$

q 가 단순히 xt−1에만 조건부인 것이 아니라 **처음 상태 x0**도 알고 있다고 가정

$$= \mathbb{E}_q\left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^{T} \log \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}\right]$$

Diffusion Model Loss(2015)

$$\mathbb{E}_q\left[-\log\frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right] = \mathbb{E}_q\left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^{T}\log\frac{q(\mathbf{x}_t|\mathbf{x}_{t-1},\mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} + \log\frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}\right]$$

$$= \mathbb{E}_q\left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^{T}\log\left(\frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}\cdot\frac{q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}\right) + \log\frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}\right]$$

Diffusion Model Loss(2015)

$$\mathbb{E}_q\left[-\log\frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right] = \mathbb{E}_q\left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^{T}\log\frac{q(\mathbf{x}_t|\mathbf{x}_{t-1},\mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} + \log\frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}\right]$$

$$= \mathbb{E}_q\left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^{T}\log\left(\frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}\cdot\frac{q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}\right) + \log\frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}\right]$$

$$= \mathbb{E}_q\left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^{T}\log\frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} + \sum_{t=2}^{T}\log\frac{q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)} + \log\frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}\right]$$

Diffusion Model Loss(2015)

$$\mathbb{E}_q\left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right] = \mathbb{E}_q\left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^{T} \log \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}\right]$$

$$= \mathbb{E}_q\left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^{T} \log \left(\frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} \cdot \frac{q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}\right) + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}\right]$$

$$= \mathbb{E}_q\left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^{T} \log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} + \sum_{t=2}^{T} \log \frac{q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)} + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}\right]$$

$$= \mathbb{E}_q\left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^{T} \log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} + \log \frac{q(\mathbf{x}_T|\mathbf{x}_0)}{q(\mathbf{x}_1|\mathbf{x}_0)} + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}\right]$$

05. Loss Function (Denoising Diffusion Probabilistic Models)

Diffusion Model Loss(2015)

$$\mathbb{E}_q\left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right] = \mathbb{E}_q\left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^{T} \log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} + \log \frac{q(\mathbf{x}_T|\mathbf{x}_0)}{q(\mathbf{x}_1|\mathbf{x}_0)} + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}\right]$$

$$= \mathbb{E}_q\left[\log \frac{q(\mathbf{x}_T|\mathbf{x}_0)}{p_\theta(\mathbf{x}_T)} + \sum_{t=2}^{T} \log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)\right]$$

05. Loss Function (Denoising Diffusion Probabilistic Models)

Diffusion Model Loss(2015)

$$\mathbb{E}_q\left[-\log\frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right] = \mathbb{E}_q\left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^{T}\log\frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} + \log\frac{q(\mathbf{x}_T|\mathbf{x}_0)}{q(\mathbf{x}_1|\mathbf{x}_0)} + \log\frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)}\right]$$

$$= \mathbb{E}_q\left[\log\frac{q(\mathbf{x}_T|\mathbf{x}_0)}{p_\theta(\mathbf{x}_T)} + \sum_{t=2}^{T}\log\frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)\right]$$

$$= \mathbb{E}_q[\underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0)\,\|\,p_\theta(\mathbf{x}_T))}_{L_T} + \sum_{t=2}^{T}\underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)\,\|\,p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}}\underbrace{- \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)}_{L_0}]$$

05. Loss Function (Denoising Diffusion Probabilistic Models)

Diffusion Model Loss(2021)

$$\mathbb{E}_q\Big[\underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p_\theta(\mathbf{x}_T))}_{L_T} + \sum_{t=2} \underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}} \underbrace{- \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)}_{L_0}\Big]$$

05. Loss Function (Denoising Diffusion Probabilistic Models)

Diffusion Model Loss(2021)

$$\boxed{\mathbb{E}_q[\underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0)\parallel p_\theta(\mathbf{x}_T))}_{L_T}} + \sum_{t=2}\underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)\parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}}\underbrace{-\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)}_{L_0}]$$

Regularization Loss에 해당
Noise 주입정도를 나타내는
Pararmeter β가 fixed되어 학습이
되지 않기에 제거

05. Loss Function (Denoising Diffusion Probabilistic Models)

Diffusion Model Loss(2021)

$$\mathbb{E}_q\Big[\underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0)\parallel p_\theta(\mathbf{x}_T))}_{L_T}+\sum_{t=2}\underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)\parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}}\underbrace{-\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)}_{L_0}\Big]$$

Regularization Loss에 해당
Noise 주입정도를 나타내는
Pararmeter β가 fixed되어 학습이
되지 않기에 제거

L0 파트는 전체적으로 봤을 때
영향력이 적기 때문에 제거

05. Loss Function (Denoising Diffusion Probabilistic Models)

Diffusion Model Loss(2021)

$$\mathbb{E}_q\Big[\underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_T|\mathbf{x}_0)\,\|\,p_\theta(\mathbf{x}_T))}_{L_T} + \boxed{\sum_{t=2}\underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)\,\|\,p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}}}\underbrace{-\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)}_{L_0}\Big]$$

이 부분을 최소화 하겠다!

Diffusion Model Loss(2021)

$$\boxed{\sum_{t=2} \underbrace{D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}}}$$

$$\downarrow$$

$$\mathbb{E}_q \left[\frac{1}{2\sigma_t^2} \| \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_\theta(\mathbf{x}_t, t) \|^2 \right]$$

KL divergence

$$
\begin{aligned}
KL(p,q) &= -\int p(x)\log q(x)dx + \int p(x)\log p(x)dx \\
&= \frac{1}{2}\log(2\pi\sigma_2^2) + \frac{\sigma_1^2 + (\mu_1-\mu_2)^2}{2\sigma_2^2} - \frac{1}{2}(1+\log 2\pi\sigma_1^2) \\
&= \log\frac{\sigma_2}{\sigma_1} + \frac{\sigma_1^2 + (\mu_1-\mu_2)^2}{2\sigma_2^2} - \frac{1}{2}
\end{aligned}
$$

주의 : 이 값은 두 분포가 정규분포라는 pdf에서 나온 식

$$
\mathbb{E}_q\left[\frac{1}{2\sigma_t^2}\|\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t,\mathbf{x}_0) - \boldsymbol{\mu}_\theta(\mathbf{x}_t,t)\|^2\right]
$$

$$N(\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$D_{KL}(p \| q) = \int p(x) \log \frac{p(x)}{q(x)} dx = \int p(x) \log p(x) dx - \int p(x) \log q(x) dx \geq 0$$

ⓑ

두 함수의 분포가 같아진다면 $D_{KL}(p \| q) = 0$

ⓐ

$$\int p(x) \log p(x) dx = \int \frac{1}{\sqrt{2\pi}\sigma_p} e^{-\frac{(x-\mu_p)^2}{2\sigma_p^2}} \left(-\frac{(x-\mu_p)^2}{2\sigma_p^2} + \log \frac{1}{\sqrt{2\pi}\sigma_p} \right) dx$$

$\frac{x-\mu_p}{\sqrt{2}\sigma_p} = t$, $\frac{1}{\sqrt{2}\sigma_p} dx = dt$, $\frac{1}{\sqrt{2\pi\sigma^2}} = z$ 치환

$$\int z \cdot e^{-t^2} (-t^2 + \log z) \sqrt{2}\sigma_p \, dt$$

$$= \frac{1}{\sqrt{\pi}} \int^{(2)} e^{-t^2} (-t^2) dt - \frac{\log(2\pi\sigma^2)}{2\sqrt{\pi}} \int^{(1)} e^{-t^2} dt$$

가우스 적분

(1) $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$

(2) $\int_{-\infty}^{\infty} x^n e^{-ax^2} dx = \frac{2(n-1)!}{2^{n/2} \cdot a^{\frac{n}{2}}} \cdot \sqrt{\frac{\pi}{a}}$

$$= -\frac{1}{2} - \frac{1}{2} \log(2\pi\sigma_p)^2 = -\frac{1}{2}(1 + \log(2\pi\sigma_p)^2)$$

ⓑ

$$-\int p(x) \log q(x) dx = -\int p(x) \log \frac{1}{\sqrt{2\pi\sigma_q^2}} \cdot e^{-\frac{(x-\mu_q)^2}{2\sigma_q^2}} dx$$

$$= -\int p(x) \log \frac{1}{\sqrt{2\pi\sigma_q^2}} dx + \int p(x) \frac{(x-\mu_q)^2}{2\sigma_q^2} dx$$

확률 모든 구간에서 확률의 합은 1. $\int p(x) dx = 1$

$$= \frac{1}{2} \log(2\pi\sigma_q)^2 + \frac{\int p(x) x^2 dx - \int 2\mu_q x p(x) dx + \mu_q^2 \int p(x) dx}{2\sigma_q^2}$$

$$= \frac{1}{2} \log(2\pi\sigma_q^2) + \frac{E_p(x^2) - 2\mu_q E_p(x) + \mu_q^2}{2\sigma_q^2}$$

$E(x) = \int x p(x) dx$ 를 이용.

$$= \frac{1}{2} \log(2\pi\sigma_q)^2 + \frac{\sigma_p^2 + \mu_p^2 - 2\mu_q\mu_p}{2\sigma_q^2} = \frac{1}{2} \log(2\pi\sigma_q^2) + \frac{\sigma_p^2 + (\mu_p - \mu_q)^2}{2\sigma_q^2}$$

ⓐ와 ⓑ에 의해

$$KL_{divergence} = \log \frac{\sigma_q}{\sigma_p} + \frac{\sigma_p^2 + (\mu_p - \mu_q)^2}{2\sigma_q^2} - \frac{1}{2}(1 + \log 2\pi\sigma_p^2)$$

$$= \log \frac{\sigma_q}{\sigma_p} + \frac{\sigma_p^2 + (\mu_p - \mu_q)^2}{2\sigma_q^2} - \frac{1}{2}$$

05. Loss Function (Denoising Diffusion Probabilistic Models)

ⓐ와 ⓑ에 의해

$$KL_{\text{divergence}} = \log \frac{\sigma_q}{\sigma_p} + \frac{\sigma_p^2 + (\mu_p - \mu_q)^2}{2\sigma_q^2} - \frac{1}{2}\left(1 + \log 2\pi \sigma_p^2\right)$$

$$= \log \frac{\sigma_q}{\sigma_p} + \frac{\sigma_p^2 + (\mu_p - \mu_q)^2}{2\sigma_q^2} - \frac{1}{2}$$

$$\mathbb{E}_q \left[\frac{1}{2\sigma_t^2} \| \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_\theta(\mathbf{x}_t, t) \|^2 \right]$$

05. Loss Function (Denoising Diffusion Probabilistic Models)

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t} \cdot x_{t-1}, \beta_t I) \longleftarrow p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_t)$$

forward

분산이 베타로 고정된 fixed값 -> 상수값

backward

ⓐ와 ⓑ에 의해

$$KL_{divergence} = \log \frac{\sigma_a}{\sigma_p} + \frac{\sigma_p^2 + (\mu_p - \mu_a)^2}{2\sigma_a^2} - \frac{1}{2}(1 + \log 2\pi \sigma_p^2)$$

$$= \log \frac{\sigma_a}{\sigma_p} + \frac{\sigma_p^2 + (\mu_p - \mu_a)^2}{2\sigma_a^2} - \frac{1}{2}$$

상수이므로 버려도 됨

즉,
q(x(t-1) | xt)의 평균 u와
pθ(xt-1|xt)의 평균 u의
차이가 최소화 되도록
pθ 네트워크를 학습시킴

$$\mathbb{E}_{t,\mathbf{x}_0,\epsilon}\left[\left\|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t)\right\|^2\right] \quad , \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

05. Loss Function (Denoising Diffusion Probabilistic Models)

$$q(x_{1:T}|x_0) = \prod_{t=1}^{T} q(x_t|x_{t-1})$$

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}\, x_{t-1}, \beta_t \mathbf{I})$$

$$x_t = \sqrt{1-\beta_t} \cdot x_{t-1} + \sqrt{\beta_t} \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I})$$

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon$$

Forward 방식.

$\underline{q(X_t | X_0)}$

$q(X_t | X_{t-1}) \sim N(X_t, \mu_{X_{t-1}}; \Sigma_{X_{t-1}}) = N(X_t, \sqrt{1-\beta_t}\, X_{t-1}, \beta_t \cdot I)$

$\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$ 라고 치환

우리는 X가 정규분포라는 가정을 하였으므로 가장 매우 중요

$X \sim N(\mu, \sigma^2)$. $x = \mu + \sigma \cdot \epsilon$, $\boxed{\epsilon \sim N(0, I)}$

$X_t = \sqrt{\alpha_t}\, X_{t-1} + \sqrt{1-\alpha_t}\, \epsilon_{t-1}$

$= \sqrt{\alpha_t \alpha_{t-1}}\, X_{t-2} + \sqrt{1-\alpha_t\alpha_{t-1}}\, \bar{\epsilon}_{t-2} = \cdots = \sqrt{\bar{\alpha}_t}\, X_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon$

$\hookrightarrow q(X_t | X_0) \sim N(X_t, \sqrt{\bar{\alpha}_t}\, X_0, (1-\bar{\alpha}_t) I)$

앞서 구한 loss에서 평균이 차이를 구하기 위해
q(xt-1 | xt, x0) 평균 계산 과 **pθ(xt-1 | xt) 평균 계산이 필요**

q(xt-1 | xt, x0) 평균

$q(x_{t-1} \mid x_t, x)$ 평균 계산

$$q(x_{t-1} \mid x_t, x_0) \sim N(x_{t-1}; \tilde{\mu}(x_t, x_0), \tilde{\beta}_t I)$$

$$q(x_{t-1} \mid x_t, x_0) = q(x_t \mid x_{t-1}, x_0) \cdot \frac{q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)}$$

$$\Rightarrow \exp\left| -\frac{1}{2}\left(\frac{(x_t - \sqrt{\alpha_t}\, x_{t-1})^2}{\beta_t} + \frac{(x_{t-1} - \sqrt{\bar{\alpha}_{t-1}}\, x_0)^2}{1-\bar{\alpha}_{t-1}} - \frac{(x_t - \sqrt{\bar{\alpha}_t}\, x_0)^2}{1-\bar{\alpha}_t} \right)\right.$$

$$= \exp\left(-\frac{1}{2}\left(\left(\frac{\alpha_t}{\beta_t} + \frac{1}{1-\bar{\alpha}_{t-1}}\right)x_{t-1}^2 - \left(\frac{2\sqrt{\alpha_t}}{\beta_t}x_t + \frac{2\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}}x_0\right)x_{t-1} + C(x_t, x_0)\right)\right)$$

$$\tilde{\mu}_t(x_t, x_0) = \frac{\frac{\sqrt{\alpha_t}}{\beta_t}x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}}x_0}{\left(\frac{\alpha_t}{\beta_t} + \frac{1}{1-\bar{\alpha}_{t-1}}\right)} = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}x_0$$

$E(X) = \int x f(x)\,dx$ 이용

$$\tilde{\beta}_t = \frac{1}{\frac{\alpha_t}{\beta_t} + \frac{1}{1-\bar{\alpha}_{t-1}}} = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t} \cdot \beta_t$$

$Var(X) = E(X^2) - E(X)^2$ 이용

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon$$

$$\hookrightarrow x_0 = \frac{1}{\sqrt{\bar{\alpha}_t}}(x_t - \sqrt{1-\bar{\alpha}_t}\, \epsilon_t)$$

$$\tilde{\mu}_t = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_t\right)$$

$\therefore$ Forward $q(x_t \mid x_0)$

$$\tilde{\mu}_t = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_t\right), \quad \tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t} \cdot \beta_t$$

pθ(xt-1 | xt) 평균

Backward

$p_\theta(x_{t-1} \mid x_t)$의 평균

$$p_\theta(x_{t-1} \mid x_t) \sim N(x_{t-1}; \mu_\theta(x_t, t), \sigma_t^2 I)$$

x_t는 입력값, α는 β에 대한 식으로 고정값

가우시안 분포 (정규분포) ϵ를 따르는 네트워크 t에 따라 학습

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t)\right)$$

다음과 표현한다.

$$x_{t-1} \sim N\left(x_{t-1}, \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t)\right), \Sigma_\theta(x, t)\right)$$

05. Loss Function (Denoising Diffusion Probabilistic Models)

(q(xt-1 | xt, x0) 평균) – (pθ(xt-1 | xt) 평균)

$$
\begin{aligned}
L_t &= \mathbb{E}_{\mathbf{x}_0,\epsilon}\Big[\frac{1}{2\|\boldsymbol{\Sigma}_\theta(\mathbf{x}_t,t)\|_2^2}\|\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t,\mathbf{x}_0)-\boldsymbol{\mu}_\theta(\mathbf{x}_t,t)\|^2\Big] \\
&= \mathbb{E}_{\mathbf{x}_0,\epsilon}\Big[\frac{1}{2\|\boldsymbol{\Sigma}_\theta\|_2^2}\|\frac{1}{\sqrt{\alpha_t}}\Big(\mathbf{x}_t-\frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_t\Big)-\frac{1}{\sqrt{\alpha_t}}\Big(\mathbf{x}_t-\frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t,t)\Big)\|^2\Big] \\
&= \mathbb{E}_{\mathbf{x}_0,\epsilon}\Big[\frac{(1-\alpha_t)^2}{2\alpha_t(1-\bar{\alpha}_t)\|\boldsymbol{\Sigma}_\theta\|_2^2}\|\boldsymbol{\epsilon}_t-\boldsymbol{\epsilon}_\theta(\mathbf{x}_t,t)\|^2\Big] \\
&= \mathbb{E}_{\mathbf{x}_0,\epsilon}\Big[\frac{(1-\alpha_t)^2}{2\alpha_t(1-\bar{\alpha}_t)\|\boldsymbol{\Sigma}_\theta\|_2^2}\|\boldsymbol{\epsilon}_t-\boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0+\sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}_t,t)\|^2\Big]
\end{aligned}
$$

Diffusion Model Probabilistic Loss(2020)

$$
\mathbb{E}_{t,\mathbf{x}_0,\epsilon}\Big[\Big\|\epsilon-\epsilon_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0+\sqrt{1-\bar{\alpha}_t}\epsilon,t)\Big\|^2\Big] \quad , \quad \epsilon\sim\mathcal{N}(\mathbf{0},\mathbf{I})
$$

06. Denoising Diffusion Probabilistic Models Code

항목	2015년 (Original Diffusion)	2020년 (DDPM, Ho et al.)
노이즈 스케줄	복잡한 방식 (β와 관련된 비선형 schedule)	간단한 선형 스케줄 사용 (β 일정 증가)
손실 함수	KL divergence 기반, 전체 ELBO 사용	간단한 **MSE loss** (노이즈 예측 기반) 사용
샘플링 방식	복잡하고 느림	개선된 샘플링 (efficiency ↑)
학습 대상	확률 분포 직접 근사 (e.g., posterior)	**노이즈 ϵ(epsilon ϵ)** 를 예측
성능	이미지 생성 품질 낮음	FID 크게 향상, 고품질 이미지 생성 가능
사용 모델	작은 네트워크	UNet 등 강력한 CNN 구조

```
def train(
    exp_name, tpu_name, bucket_name_prefix, model_name='unet2d16b2c112244', dataset='celebahq256',
    optimizer='adam', total_bs=64, grad_clip=1., lr=0.00002, warmup=5000,
    num_diffusion_timesteps=1000, beta_start=0.0001, beta_end=0.02, beta_schedule='linear', loss_type='noisepred',
    dropout=0.0, randflip=1, block_size=1,
    tfds_data_dir='tensorflow_datasets', log_dir='logs'
):
  region = utils.get_gcp_region()
  tfds_data_dir = 'gs://{}-{}/{}'.format(bucket_name_prefix, region, tfds_data_dir)
  log_dir = 'gs://{}-{}/{}'.format(bucket_name_prefix, region, log_dir)
  kwargs = dict(locals())
  ds = datasets.get_dataset(dataset, tfds_data_dir=tfds_data_dir)
  tpu_utils.run_training(
    date_str='9999-99-99',
    exp_name='{exp_name}_{dataset}_{model_name}_{optimizer}_bs{total_bs}_lr{lr}w{warmup}_beta{beta_start}-{beta_end}-{bet
      **kwargs),
    model_constructor=lambda: Model(
      model_name=model_name,
      betas=get_beta_schedule(
        beta_schedule, beta_start=beta_start, beta_end=beta_end, num_diffusion_timesteps=num_diffusion_timesteps
      ),
      loss_type=loss_type,
      num_classes=ds.num_classes,
      dropout=dropout,
      randflip=randflip,
      block_size=block_size
    ),
    optimizer=optimizer, total_bs=total_bs, lr=lr, warmup=warmup, grad_clip=grad_clip,
    train_input_fn=ds.train_input_fn,
    tpu=tpu_name, log_dir=log_dir, dump_kwargs=kwargs
  )
```

공식코드
(https://github.com/hojonathanho/diffusion)

06. Denoising Diffusion Probabilistic Models Code

공식코드
(https://github.com/hojonathanho/diffusion)

```
def train(
    exp_name, tpu_name, bucket_name_prefix, model_name='unet2d16b2c112244', dataset='celebahq256',
    optimizer='adam', total_bs=64, grad_clip=1., lr=0.00002, warmup=5000,
    num_diffusion_timesteps=1000, beta_start=0.0001, beta_end=0.02, beta_schedule='linear', loss_type='noisepred',
    dropout=0.0, randflip=1, block_size=1,
    tfds_data_dir='tensorflow_datasets', log_dir='logs'
):
  region = utils.get_gcp_region()
  tfds_data_dir = 'gs://{}-{}/{}'.format(bucket_name_prefix, region, tfds_data_dir)
  log_dir = 'gs://{}-{}/{}'.format(bucket_name_prefix, region, log_dir)
  kwargs = dict(locals())
  ds = datasets.get_dataset(dataset, tfds_data_dir=tfds_data_dir)
  tpu_utils.run_training(
    date_str='9999-99-99',
    exp_name='{exp_name}_{dataset}_{model_name}_{optimizer}_bs{total_bs}_lr{lr}w{warmup}_beta{beta_start}-{beta_end}-{bet
      **kwargs),
    model_constructor=lambda: Model(
      model_name=model_name,
      betas=get_beta_schedule(
        beta_schedule, beta_start=beta_start, beta_end=beta_end, num_diffusion_timesteps=num_diffusion_timesteps
      ),
      loss_type=loss_type,
      num_classes=ds.num_classes,
      dropout=dropout,
      randflip=randflip,
      block_size=block_size
    ),
    optimizer=optimizer, total_bs=total_bs, lr=lr, warmup=warmup, grad_clip=grad_clip,
    train_input_fn=ds.train_input_fn,
    tpu=tpu_name, log_dir=log_dir, dump_kwargs=kwargs
  )
```

so that $L_T \approx 0$. We set $T = 1000$ without a sweep, and we chose a linear
m $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$.

06. Denoising Diffusion Probabilistic Models Code

Algorithm 1 Training

1: **repeat**
2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
3: $t \sim \mathrm{Uniform}(\{1, \ldots, T\})$
4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: Take gradient descent step on
$$\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t) \right\|^2$$
6: **until** converged

```python
def train_fn(self, x, y):
  B, H, W, C = x.shape
  if self.randflip:
    x = tf.image.random_flip_left_right(x)
    assert x.shape == [B, H, W, C]
  t = tf.random_uniform([B], 0, self.diffusion.num_timesteps, dtype=tf.int32)
  losses = self.diffusion.p_losses(
    denoise_fn=functools.partial(self._denoise, y=y, dropout=self.dropout), x_start=x, t=t)
  assert losses.shape == t.shape == [B]
  return {'loss': tf.reduce_mean(losses)}
```

공식코드
(https://github.com/hojonathanho/diffusion)

06. Denoising Diffusion Probabilistic Models Code

Algorithm 2 Sampling

1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \ldots, 1$ **do**
3: $\quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
4: $\quad \mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$
5: **end for**
6: **return** $\mathbf{x}_0$

```
def samples_fn(self, dummy_noise, y):
  return {
    'samples': self.diffusion.p_sample_loop(
      denoise_fn=functools.partial(self._denoise, y=y, dropout=0),
      shape=dummy_noise.shape.as_list(),
      noise_fn=tf.random_normal
    )
  }
```

공식코드
(https://github.com/hojonathanho/diffusion)

THANK YOU!

이건, 주현빈